Enterprise Web Integration Architecture and Master Prompt Specification for Formula 1 Race Intelligence

The integration of enterprise Business Intelligence (BI) dashboards into web applications provides an effective mechanism for delivering domain-specific analytics to non-technical stakeholders, team strategists, and broadcast journalists. Within motorsport analytics, raw telemetry and historical race logs require visual synthesis to highlight performance deltas, mechanical reliability, and strategy optimization. The Formula 1 Race Intelligence analytical suite synthesizes 17 seasons of racing data (2010–2026), incorporating 6,464 individual race outcome logs across 29 global circuits and 10 competing constructors.
To translate this Tableau analytical suite into an application hosted via micro-frameworks such as Flask or static production hosts, a structured architectural prompt is required for AI-driven code generation engines like Google AI Studio. This report details the data architecture, visual design system, embedding protocols, and an production-ready master prompt optimized for automated web buildout.

Analytical Architecture and Multi-Dimensional Telemetry

The data foundation powering the analytical engine spans driver performance trajectories, constructor points progression, tyre degradation models, spatial Grand Prix distributions, and mechanical reliability metrics. The underlying Tableau Public architecture structures these metrics into eight visual modules and a seven-tab analytical story, providing both interactive dashboard filtering and sequential story-driven insight delivery.

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Formula 1 Race Intelligence Dataset (6,464 Records | 17 Seasons | 29 Circuits | 10 Teams)
 ├── Spatial Analytics: Global Driver Distribution (13 Countries Mapped)
 ├── Performance Metrics: Driver Points Leaderboard (Hamilton: 31,147, Verstappen: 31,126)
 ├── Constructor Tracking: Team Performance Ranking (Red Bull Racing: 8,006 pts lead)
 ├── Strategic Stints: Tyre Compound Usage (Soft: 31.52%, Medium: 30.78%, Hard: 30.36%)
 ├── Circuit Logs: Country-Wise Participation Treemap (Bahrain: 2,168 records)
 └── Reliability Metrics: DNF Breakdown & Driver Reliability Index
```

Dashboard Visualization Catalog

Visualization Module	Chart Type	Key Data Points & Metric Values	Operational Analytical Function
Global Driver Distribution	Geographic Map	USA (400), Spain (328), Monaco (328), Canada (308), Brazil (308), Singapore (288), Mexico (160), Japan (140), Netherlands (140), Saudi Arabia (140), Azerbaijan (140), Australia (140), China (40)	Maps driver talent pools and national representation across international Grand Prix host countries.
Driver Reliability Analysis	Stacked Bar Chart	Categorized Did Not Finish (DNF) counts (engine, mechanical, brake failures, collisions) across 19 drivers including Lewis Hamilton, Max Verstappen, Charles Leclerc, Lando Norris, Fernando Alonso, and Carlos Sainz	Evaluates driver reliability trends and mechanical attrition rates over multi-season regulation shifts.
Driver Performance Leaderboard	Vertical Bar Chart	Lewis Hamilton (31,147), Max Verstappen (31,126), Charles Leclerc (27,673), Lando Norris (26,408), Yuki Tsunoda (14,438), Esteban Ocon (14,205), Carlos Sainz (11,004), Fernando Alonso (1,353)	Tracks cumulative career points and benchmarks driver scoring trajectories.
Team Performance Ranking	Horizontal Bar Chart	Red Bull Racing (8,006), Mercedes, Ferrari, McLaren, Alpine, Aston Martin, Williams, RB (AlphaTauri), Haas, Kick Sauber	Ranks constructors based on aggregate points accumulated over the 17-season window.
Team Championship Analysis	Vertical Bar Chart	Comparative points distribution across all 10 active constructor entities	Compares season-over-season championship scoring across top-tier and mid-grid constructor teams.

Visualization Module	Chart Type	Key Data Points & Metric Values	Operational Analytical Function
Tyre Compound Usage	Pie / Donut Chart	Soft (31.52%), Medium (30.78%), Hard (30.36%), Intermediate (7.34%)	Details stint compound allocation strategies across diverse track temperatures and circuit profiles.
Country-Wise Participation	Treemap	Bahrain (2,168), USA (400), Hungary (328), Monaco (328), Spain (328), Brazil (308), Canada (308), UAE (308), Singapore (288), plus Italy, Austria, Qatar, UK	Visualizes total race entry logs grouped by host country venues.
Podium Achievements	Packed Bubble Chart	Podium density and distribution by driver and team (highlighting volume clusters for Ferrari and Mercedes)	Displays top-three finish concentrations per driver and team combination.

Stakeholder Use-Case Alignment

The operational utility of the web application relies on serving three distinct user personas identified in the system requirements, translating analytical outputs into actionable domain decisions:

- **F1 Team Performance Strategist**: Benchmarks mid-tier team points trajectories against top constructors like Red Bull Racing and Mercedes, evaluates tyre compound degradation (Soft, Medium, Hard, Intermediate) across circuit types, and evaluates international driver talent pools to optimize recruitment and race strategy.
- **Motorsport Broadcast Analyst**: Accesses real-time leaderboards, historical Grand Prix host nation data, and podium volume bubbles to construct pre-race infographics, live commentary statistics, and post-race media analysis.
- **FIA Technical Regulation Analyst**: Analyzes mechanical failure patterns (engine, brake, structural) and driver reliability scores across 17 seasons to determine whether historical technical regulation iterations successfully improved car completion rates and reduced on-track collisions.

Front-End UI/UX Design System and Layout Architecture

High-contrast, telemetry-inspired visual interfaces are essential for motorsport analytics platforms. The visual theme utilizes an aggressive dark background palette complemented by Formula 1 racing red accents, structured card components, and clean typography.

Visual Design Tokens

Token Category	Specification / Hex Code	Interface Implementation
Primary Background	#0B0E14 (Deep Slate Black)	Core page body background and hero container base.
Secondary Background	#151921 (Carbon Gray)	Card containers, sticky navigation bar, and interactive grid widgets.
Container Border	#262C38 (Muted Steel)	Borders for dashboard frames, stat counters, and live ticker banners.
Primary Accent	#E10600 (F1 Racing Red)	Buttons, badges, active navigation indicators, and key numerical highlights.
Hover Accent	#FF1801 (Crimson Glow)	Interactive hover states, active tickers, and call-to-action highlights.
Primary Text	#FFFFFF (Pure White)	Headings, hero display titles, primary KPI numbers, and navigation text.
Muted Text	#94A3B8 (Cool Slate)	Section subtitles, body copy, chart descriptions, and metadata labels.
Interface Typography	Inter, system-ui, sans-serif	Primary body text, navigation elements, grid labels, and narrative descriptions.
Display Typography	Formula1 Display, Titillium Web	Hero headlines, section titles, card headers, and stat strip numbers.

Page Layout Hierarchy

The web interface follows a linear, eight-stage section architecture designed to lead the user from broad executive metrics down to granular embedded visualization viewports:

1. **Sticky Navigation Header**: A fixed dark top bar (#151921) featuring an italicized red "F1" badge alongside uppercase "RACE INTELLIGENCE" branding. Navigation links comprise HOME, PROJECT, DASHBOARD, STORY, GALLERY, INSIGHTS, DRIVERS, and a highlighted red pill button for CONTACT.
2. **Hero Section**: Contains a live status badge reading "LIVE RACE ANALYTICS PLATFORM", a high-impact headline "FORMULA 1 RACE INTELLIGENCE DASHBOARD", and narrative copy summarizing the 17-season telemetry dataset. Call-to-action buttons provide direct anchor navigation to "VIEW DASHBOARD" and "EXPLORE STORY".
3. **Primary KPI Stat Strip**: A four-card metric bar highlighting high-level scope: 75+ Seasons Analyzed, 1K+ Races Covered, 20 Active Drivers, and 10 Constructors.
4. **Secondary Operational Stat Strip**: A secondary four-card strip displaying operational counts: 897+ Total Race Entries, 33 Championship Titles, 372\text{ km/h} Top Speed, and 15+ KPIs Visualized.
5. **Project Overview Container**: A two-column analytical narrative titled *"Transforming F1 Data Into Championship Intelligence"* accompanied by descriptive copy, visual tags (Data

Analytics, Tableau, Race Strategy, Visualization, Championship, F1 Dataset), and feature checklists covering circuit telemetry, lap times, pit strategy, and championship forecast models.

6. **Intelligence Dimensions & Telemetry Ticker**: Features a continuously scrolling horizontal telemetry ticker displaying real-time headlines (e.g., *"Monaco GP: Fastest Lap 1:12.492"*, *"Ferrari leads avg pit stop speed in 2024"*, *"Red Bull Racing leads Constructor Championship"*) positioned directly above a 6-card grid:
    - *Circuit Analysis*: Track-by-track mapping of 24 Grand Prix locations comparing high-speed vs. technical downforce setups.
    - *Lap Time Breakdown*: Sector-by-sector driver telemetry delta analysis at millisecond resolution.
    - *Tire Strategy*: Stint optimization models across 5 compound categories (Soft, Medium, Hard, Inter, Wet).
    - *Driver Head-to-Head*: Comparative performance deltas between teammates across qualifying pace, race speed, and DNF rates.
    - *Points Progression*: Animated multi-season championship points trajectory tracking.
    - *Weather & Safety Cars*: Strategic impact analysis evaluating rain transitions, Virtual Safety Cars, and full Safety Car deployments.
7. **Embedded Tableau Dashboard Viewport**: An interactive iframe frame rendering the complete 8-chart Tableau Public dashboard workbook.
8. **Embedded Tableau Story Viewport**: An interactive frame embedding the 7-tab Tableau Public story presentation.

Technical Web Embedding, API Parameters, and Breakpoint Engineering

Embedding Tableau Public workbooks cleanly into custom web templates requires specific URL parameter configuration to strip native site chrome while maintaining full interactive behavior (tooltips, mark selection, dropdown filters, and story tab navigation).

Tableau URL Parameters

Parameter	Optimized Value	Technical Behavior & Purpose
:embed	yes	**Mandatory.** Hides top Tableau Server/Public site header and navigation controls, blending the viz into the webpage.
:showVizHome	no	Prevents the default Tableau Public author profile frame from appearing within the embedded container.
:toolbar	n	Hides the native Tableau footer toolbar to eliminate visual clutter.
:display_count	yes	Displays cumulative views when interacting with

Parameter	Optimized Value	Technical Behavior & Purpose
		embedded visual elements.
:device	desktop	Forces desktop visual layout rendering across large screens.

The target embed links for the project are:

- **Dashboard Embed URL**: https://public.tableau.com/views/F1RACEINTELLIGENCEDASHBOARD/Dashboard?:embed=yes&:showVizHome=no&:toolbar=n&:display_count=yes
- **Story Embed URL**: https://public.tableau.com/views[span_9](start_span)[span_9](end_span)[span_12](start_span)[span_12](end_span)/F1RACEStory/Story1?:embed=yes&:showVizHome=no&:toolbar=n&:display_count=yes

Responsive Sizing and Dynamic Resizing

To maintain visual integrity across devices without introducing dual scrollbars, JavaScript logic dynamically detects container width (offsetWidth) and assigns appropriate frame height values before loading Tableau's native JavaScript library (viz_v1.js).
For viewports wider than 800 pixels (desktop display), the container height is locked to 927px with 100% width. For viewports between 501 and 800 pixels (tablet display), the height remains 927px. For mobile viewports of 500 pixels or fewer, the container height expands dynamically to 2127px to accommodate stacked vertical card layouts.

Production-Grade Master Prompt for Google AI Studio

The master prompt below is optimized for direct submission into Google AI Studio (or Gemini Code Assist). It instructs the AI to generate a single-file, production-ready web application incorporating all visual, structural, and embedding requirements.

SYSTEM ROLE & GOAL

You are a Principal Frontend Developer and Data Visualization Engineer. Build a complete, single-file, production-ready web application using HTML5, Tailwind CSS (via CDN), and JavaScript that embeds a Tableau Public Formula 1 Analytics platform. The web application must feature an F1 motorsport dark theme, high-impact KPI stat counters, interactive analytical cards, a live scrolling telemetry banner, and dynamic responsive iframe wrappers for embedded Tableau Public dashboards and stories.

TECH STACK REQUIREMENTS

1. Architecture: Single-file HTML5 document (index.html).
2. CSS Framework: Tailwind CSS v3 via CDN (https://cdn.jsdelivr.net/npm/tailwindcss@3.4.1/dist/tailwind.min.js). Include Lucide Icons via CDN (https://unpkg.com/lucide@latest) for telemetry icons.
3. Design Palette: F1 Motorsport Dark Theme
    - Base Background: #0B0E14
    - Card/Container Base: #151921

    - Borders: #262C38
    - Primary Accent: #E10600 (F1 Racing Red)
    - Accent Hover: #FF1801
    - Text Primary: #FFFFFF
    - Text Secondary: #94A3B8
    - Font Family: Inter, Titillium Web, sans-serif
4. Tableau Public Embed Sources:
    - Dashboard Viz: https://public.tableau.com/views/F1RACEINTELLIGENCEDASHBOARD/Dashboard
    - Story Viz: https://public.tableau.com/views/F1RACEStory/Story1
    - Author: Gowtham reddy Thumu

REQUIRED SECTION SPECIFICATIONS

1. NAVIGATION BAR (Fixed Top Header)

- Background: #151921 with bottom border #262C38.
- Left Branding: An italicized red badge (#E10600) containing "F1" in bold white text, followed by "RACE INTELLIGENCE" in uppercase bold text.
- Nav Links: HOME, PROJECT, DASHBOARD, STORY, GALLERY, INSIGHTS, DRIVERS. Hover state: text-red-500.
- Right CTA Button: "CONTACT" styled as a solid red (#E10600) pill button with red shadow hover.

2. HERO SECTION

- Background: #0B0E14 with a subtle red radial background glow.
- Status Badge: Bordered pill badge with red dot: "LIVE RACE ANALYTICS PLATFORM".
- Display Headline: Large bold display font: "FORMULA 1 RACE INTELLIGENCE DASHBOARD".
- Subtitle Copy: "Deep-dive analytics into Formula 1 Championship data, from driver performance metrics and team strategy to circuit analysis and historical championship trends. Powered by Tableau visualization intelligence."
- Action Buttons:
    - Button 1: "VIEW DASHBOARD" (Solid Red #E10600, smooth scroll anchor to #dashboard).
    - Button 2: "EXPLORE STORY" (Outline button with white border, smooth scroll anchor to #story).

3. DUAL STAT COUNTER STRIPS

- Primary Stat Strip (4-Card Grid):
    - Card 1: 75+ Seasons Analyzed
    - Card 2: 1K+ Races Covered
    - Card 3: 20 Active Drivers
    - Card 4: 10 Constructors
- Operational Stat Strip (4-Card Dark Accent Grid):
    - Card 1: 897+ Total Race Entries

    - Card 2: 33 Championship Titles
    - Card 3: 372 km/h Top Speed
    - Card 4: 15+ KPIs Visualized

4. PROJECT OVERVIEW CONTAINER (ID: project)

- Heading: "Transforming F1 Data Into Championship Intelligence"
- Narrative Paragraph: Detail how the platform transforms 17 seasons of racing telemetry (2010–2026) comprising 6,464 race outcome logs across 29 circuits into strategic insights for team strategists, broadcast analysts, and FIA technical compliance analysts.
- Tech Stack Tags: Data Analytics, Tableau, Race Strategy, Visualization, Championship, F1 Dataset.
- Feature Checklist (Checkmark icons):
    - Global Race Overview: Circuit-by-circuit telemetry across 24 Grand Prix host venues.
    - Driver Performance Metrics: Lap times, qualifying pace, race pace, overtaking deltas.
    - Constructor Championship Battle: Team points progression and reliability models.
    - Pit Stop Strategy Analysis: Tire degradation models and optimal stint windows.
    - Championship Success Forecast: Historical trends and statistical probability models.
- Side Graphic: Stylized SVG race circuit vector illustration with colored telemetry position markers.

5. LIVE TELEMETRY TICKER & INTELLIGENCE DIMENSIONS (ID: insights)

- Telemetry Ticker: Infinite horizontally scrolling black bar with top/bottom red borders. Ticker text elements:
    - "⚡ Monaco GP: Fastest Lap 1:12.492"
    - "⚡ Ferrari leads avg pit stop speed in 2024"
    - "⚡ Red Bull Racing leads Constructor Championship"
    - "⚡ Lewis Hamilton: 31,147 Career Points Logged"
    - "⚡ Soft Compound Usage: 31.52% Stint Share"
- Grid Title: "Analytical Intelligence Dimensions"
- 6-Card Grid Layout (Card background #151921, border #262C38, hover transition up with red accent glow):
    1. Circuit Analysis: Track performance data across 24 Grand Prix circuits from Monaco to Monza.
    2. Lap Time Breakdown: Sector-by-sector driver delta analysis at millisecond resolution.
    3. Tire Strategy: Degradation rates and optimal stint windows across 5 compound categories.
    4. Driver Head-to-Head: Teammate and rival comparisons across qualifying deltas, pace, and DNFs.
    5. Points Progression: Season-long interactive points trajectory visualization.
    6. Weather & Safety Cars: Impact models for weather shifts, Virtual Safety Cars, and full SC deployments.

6. EMBEDDED TABLEAU DASHBOARD SECTION (ID: dashboard)

- Container Title: "Race Intelligence Dashboard"
- Subtitle: "Explore 17 seasons of driver performance leaderboards, team rankings, and tyre strategies."
- Code Implementation: Insert the standard Tableau Public JavaScript embed wrapper targeting F1RACEINTELLIGENCEDASHBOARD/Dashboard.

7. EMBEDDED TABLEAU STORY SECTION (ID: story)

- Container Title: "Formula 1 Race Intelligence: Story Narrative"
- Subtitle: "Step-by-step guided story narrative from global overview to championship outcomes."
- Code Implementation: Insert the standard Tableau Public JavaScript embed wrapper targeting F1RACEStory/Story1.

8. FOOTER SECTION

- Content: Brand description, F1 logo badge, quick nav links, Tableau Public direct links, GitHub project link, and evaluation submission badge: "Project Status: Ready for Evaluation | Author: Gowtham reddy Thumu".

EMBED CODE PATTERN TO INCLUDE

Ensure both embed containers use this exact JavaScript structure:html <noscript>   </noscript> <object class="tableauViz" style="display:none;"> <param name="host_url" value="https%3A%2F%2Fpublic.tableau.com%2F" /> <param name="embed_code_version" value="3" /> <param name="site_root" value="" /> <param name="name" value="F1RACEINTELLIGENCEDASHBOARD/Dashboard" /> <param name="tabs" value="no" /> <param name="toolbar" value="no" /> <param name="showVizHome" value="no" /> </object> <script type="text/javascript"> var divElement = document.getElementById('viz_dashboard_container'); var vizElement = divElement.getElementsByTagName('object')[0]; if (divElement.offsetWidth > 800) { vizElement.style.width = '100%'; vizElement.style.height = '927px'; } else if (divElement.offsetWidth > 500) { vizElement.style.width = '100%'; vizElement.style.height = '927px'; } else { vizElement.style.width = '100%'; vizElement.style.height = '2127px'; } var scriptElement = document.createElement('script'); scriptElement.src = 'https://public.tableau.com/javascripts/api/viz_v1.js'; vizElement.parentNode.insertBefore(scriptElement, vizElement); </script>

STYLING & QUALITY DIRECTIVES
- Set `html { scroll-behavior: smooth; }` for clean anchor

```
transitions.
- Ensure strict responsive scaling across desktop, tablet, and mobile
views.
- Provide COMPLETE, clean HTML/JS output without placeholder code or
missing sections.
```

Deployment Architecture and Verification Protocols

To host the generated application, developers can execute a Flask micro-service deployment or serve the asset directly via static Web servers (GitHub Pages, Vercel, or Netlify).

Flask Web Server Setup

In a Flask application structure, the output generated by Google AI Studio is saved as index.html within the templates/ folder:

```
from flask import Flask, render_template

app = Flask(__name__)

@app.route('/')
def index():
    """Renders the Formula 1 Race Intelligence web application
template."""
    return render_template('index.html')

if __name__ == '__main__':
    # Launch local development server on port 5000
    app.run(debug=True, host='0.0.0.0', port=5000)
```

Technical Evaluation Criteria

Evaluation Dimension	Metric / Standard	Verification Standard
Data Integrity	Matches 6,464 race logs across 17 seasons, including exact scoring metrics (Hamilton: 31,147 pts; Verstappen: 31,126 pts; Red Bull Racing: 8,006 pts).	Verify chart tooltip values in embedded frames against source dataset records.
Embed Responsiveness	Dynamic viewport scaling across Desktop (>800\text{px}), Tablet (501\text{--}800\text{px}), and Mobile (\le 500\text{px}) breakpoints.	Test browser inspection window across varying responsive resolutions.
Interactive Filtering	Active parameter filtering across DNF Reason, Year of	Validate interactive mark selection and parameter

Evaluation Dimension	Metric / Standard	Verification Standard
	Race Date, and Top N Parameter (set to 10) inside iframe containers.	dropdown behavior.
Design Consistency	Full compliance with dark motorsport UI tokens (#0B0E14 base, #E10600 accent red, #151921 container background).	Perform visual UI review against F1 aesthetic guidelines.
Submission Integrity	Inclusion of valid Tableau Public workbook URLs, GitHub repository links, and active demo links in footer and banner.	Confirm external URL link routing and mentor evaluation banner links.

Works cited

1. Embedding API - Tableau, https://www.tableau.com/developer/tools/embedding-api 2. A Guide to Embedding Tableau # 2: Embed Examples, https://www.flerlagetwins.com/2023/05/embedding-2.html 3. Tailwind CSS Dark Mode | Free Preline UI Components, https://preline.co/docs/dark-mode.html 4. Embed Code Parameters for Views - Tableau Help, https://help.tableau.com/current/pro/desktop/en-us/embed_list.htm 5. Embedding Tableau Public Dashboard into the web application, https://dev.to/prashantkamble93/embedding-tableau-public-dashboard-into-the-web-application-1pmf 6. Embed Dashboards - Tableau Help, https://help.tableau.com/current/pro/desktop/en-us/embed_dashboard.htm